Skills Summary

Selecting Ratio Representations for Comparisons

Use this reference while completing your worksheet or when studying.

Comparing with Unit Rates

Rule: When two offers or speeds use **different** amounts, divide to get each **unit rate** (per one pound, per one hour), then compare the unit rates directly. Choose this tool when the question says "per" or "which is cheaper/faster."

Example: Store P sells 4 lb of apples for 10 dollars; store Q sells 3 lb for 9 dollars. Which is cheaper per pound?
Step 1. Different amounts, so compare unit prices: $10 \div 4 = 2.5$ dollars per pound and $9 \div 3 = 3$ dollars per pound.
Step 2. Comparing the unit prices, store P costs less per pound: $\mathbf{2.5 < 3}$.

Try it: Store R sells 5 lb for 12 dollars; store S sells 4 lb for 10 dollars. Write the comparison of their price-per-pound fractions with < or >.

check: $\frac{12}{5} < \frac{10}{4}$

Scaling to a Common Amount

Rule: A ratio table or double number line lets you scale two ratios to the **same** amount (same time, same servings) and compare the other quantity. Choose this tool when both ratios can reach a shared amount easily.

Example: Ava walks 8 km every 2 hours; Ben walks 9 km every 3 hours. Who goes farther in 6 hours?
Step 1. Scale both to 6 hours: Ava $8 \times 3 = 24$ km, Ben $9 \times 2 = 18$ km.
Step 2. Over the same 6 hours, Ava goes farther: $\mathbf{24 > 18}$.

Try it: Cara bikes 10 km every 2 hours; Dan bikes 12 km every 4 hours. Compare their km-per-hour fractions with < or >.

check: $\frac{10}{2} > \frac{12}{4}$

(!) Watch out: Never compare raw numbers from different-sized batches — 210 km "beats" 140 km until you notice the times differ. Scale to a common amount (or a unit rate) first.

Splitting a Total with a Tape Diagram

Rule: When a **total** is shared in a ratio, draw a tape: one box per ratio part. Add the parts, divide the total by the number of boxes to value one box, then multiply. Choose this tool when the problem gives a total (or a difference).

Example: A 45-marble bag is split in the ratio 2:7. How many marbles is the larger share?
Step 1. The tape has $2 + 7 = 9$ boxes for 45 marbles, so one box is $45 \div 9 = 5$ marbles.
Step 2. The larger share has 7 boxes: $7 \times 5 = 35$, so it is **35** marbles.

Try it: A 36-sticker sheet is split in the ratio 5:4. How many stickers is the larger share?

check: **20**

Key Vocabulary

Unit rate: the amount per one unit

Tape diagram: equal boxes splitting a total by ratio parts

Common amount: a shared quantity two ratios are scaled to

Representation: the tool you choose to model a ratio